Noise Filtering

Filtering

Noisy Image

Filtered Image

Noise Filtering

Filtering

Noisy Image

Filtered Image

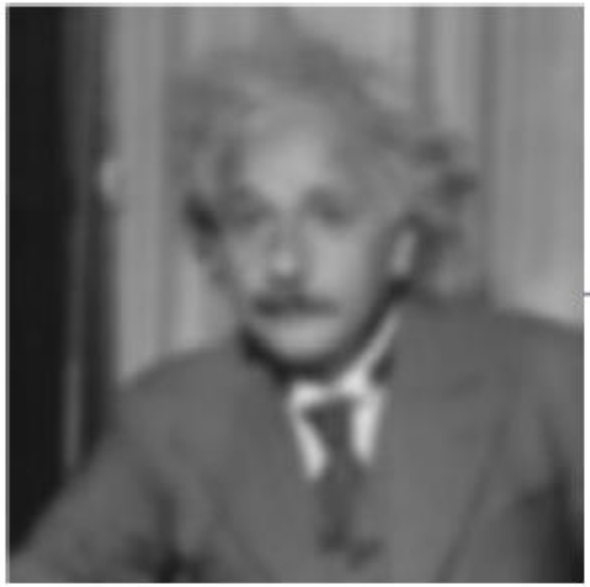
Image Enhancement

Contrast enhancement

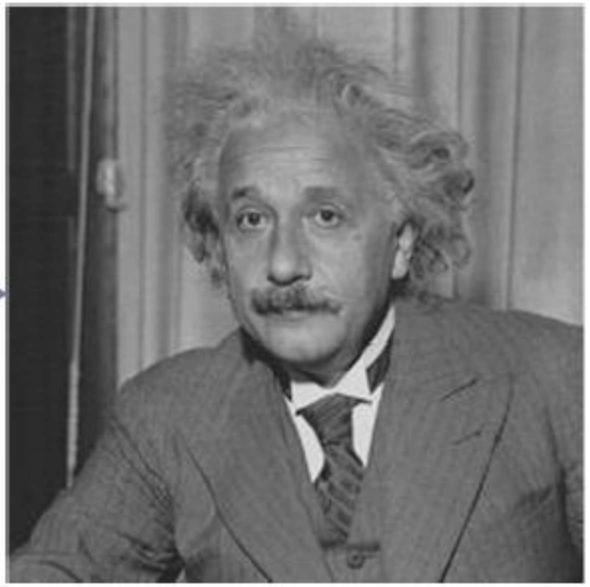
Low contrast image

Enhanced image

Deblurring



Deblurring



Blurred image

Deblurred image

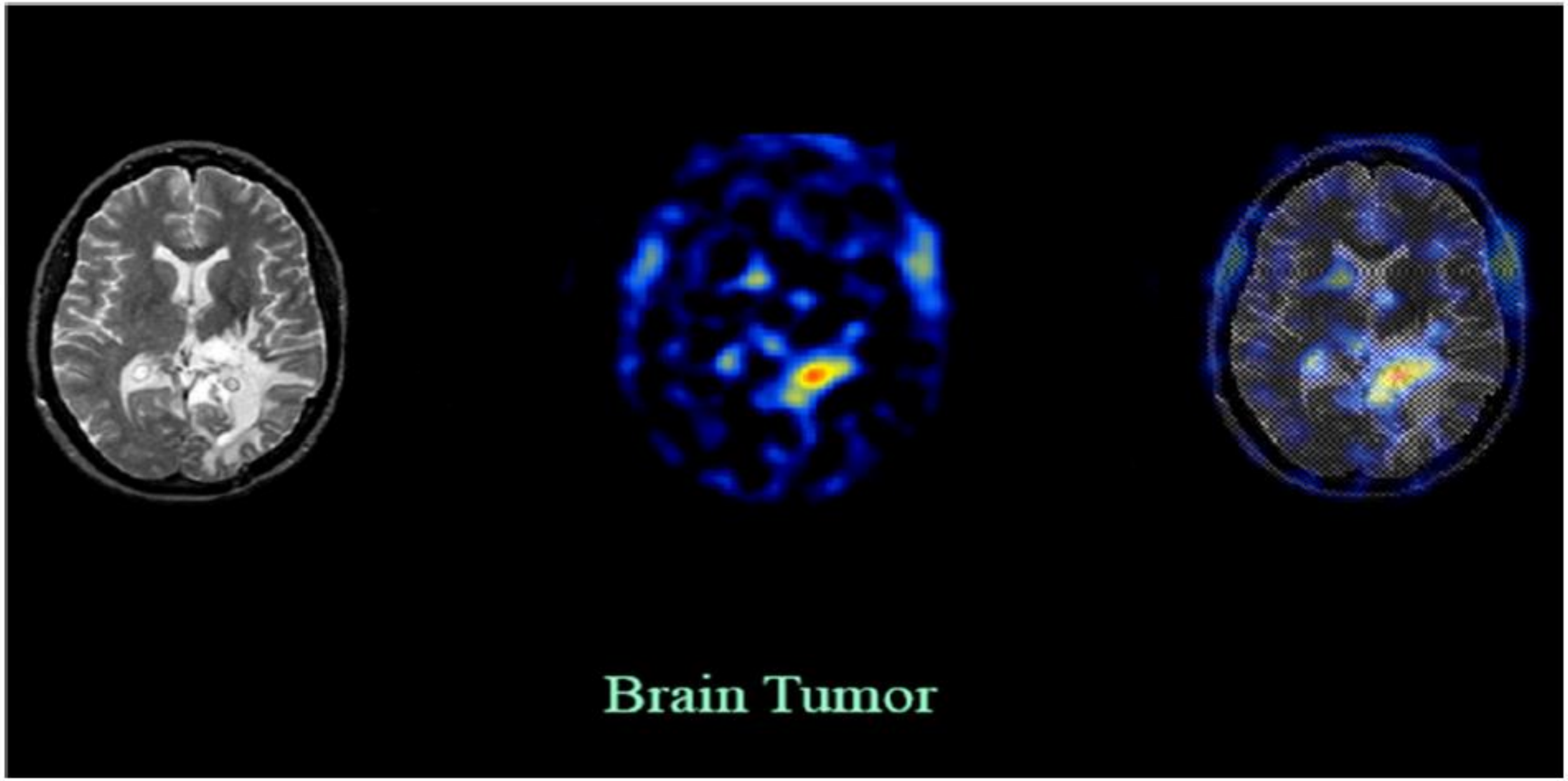
Deblurring

Deblurring

Blurred image

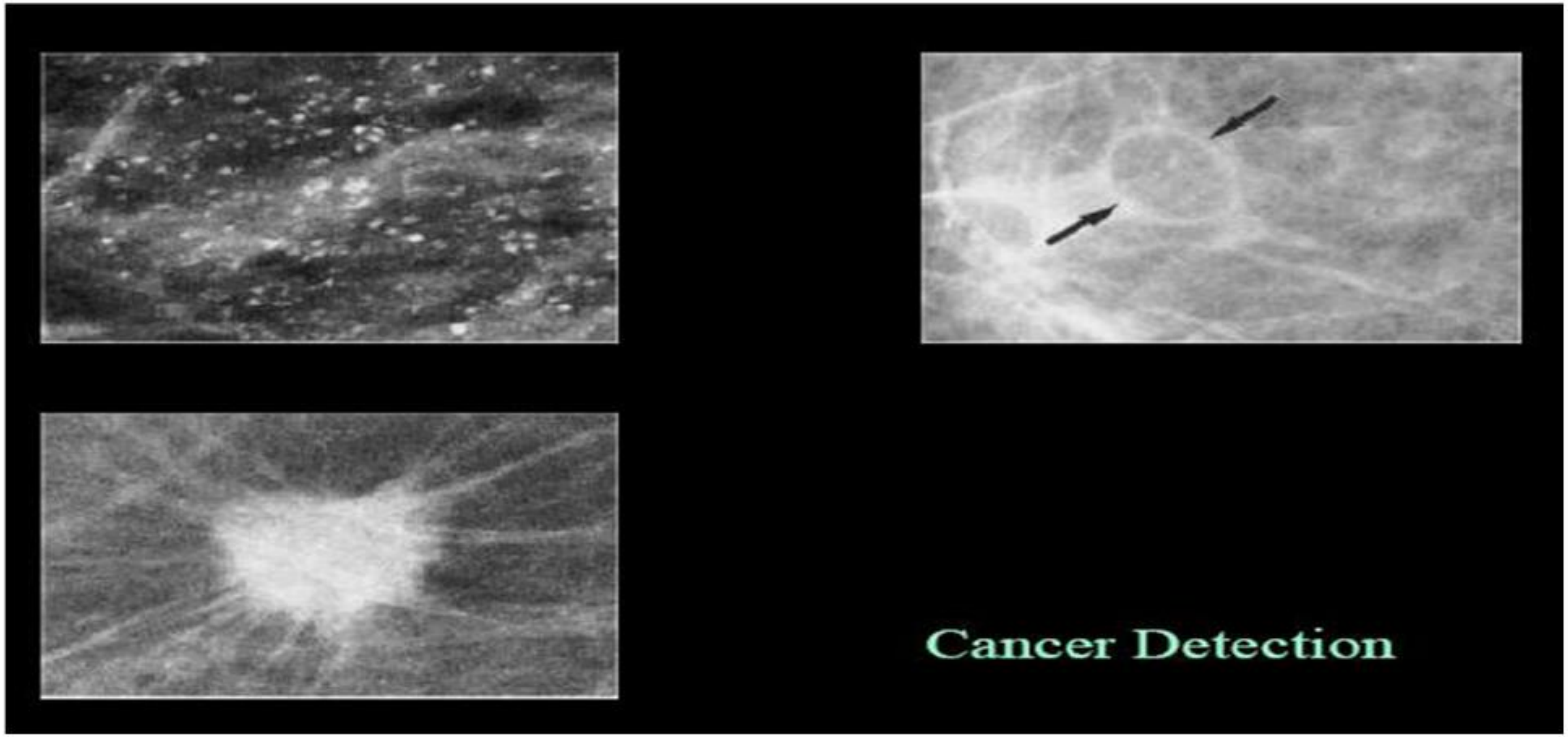
Deblurred image

Medical Imaging



CT Scan

Medical Imaging



CT Scan

Remote Sensing

Satellite Image, Kolkata

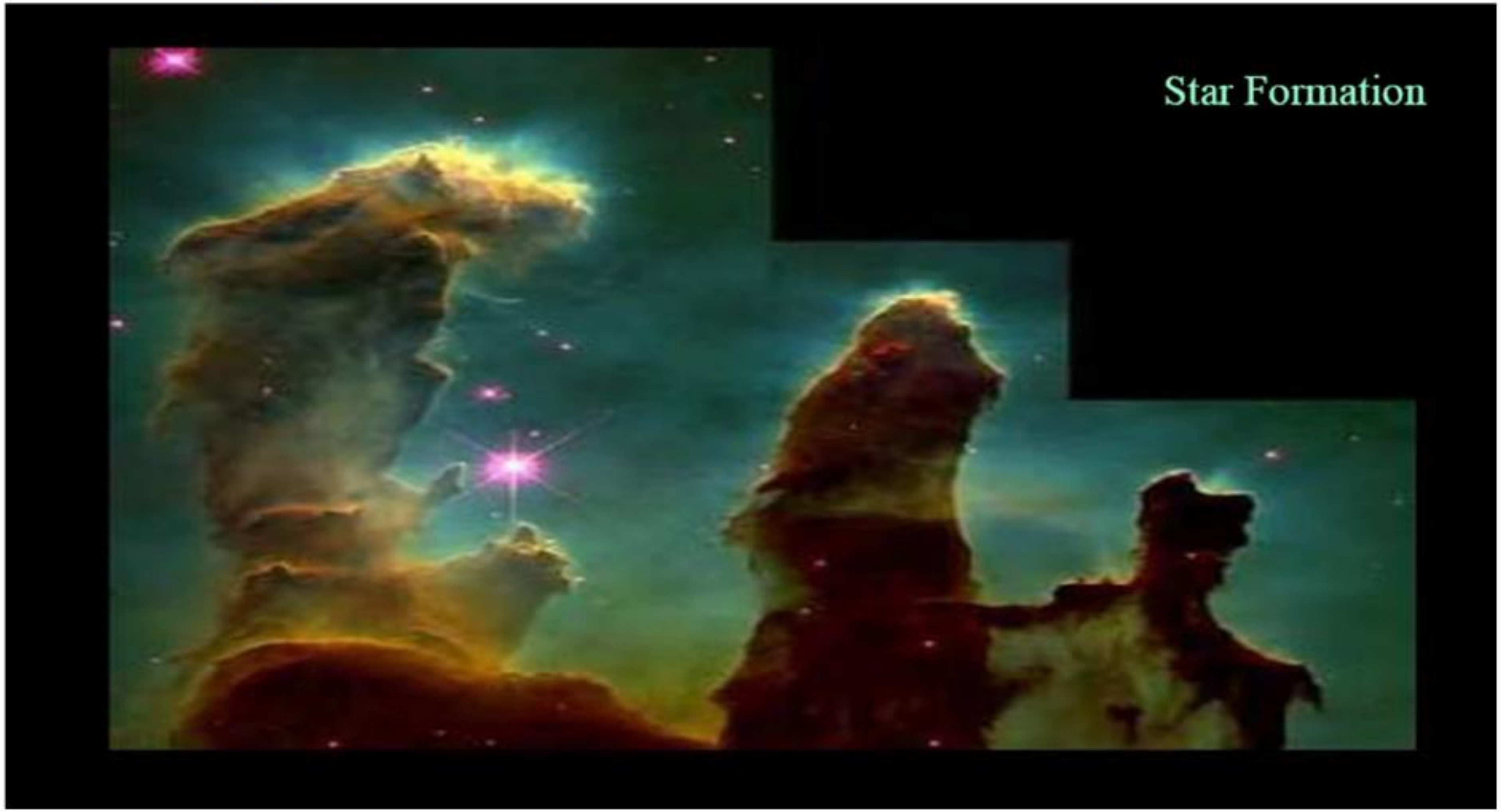
Remote Sensing

Terrain Mapping

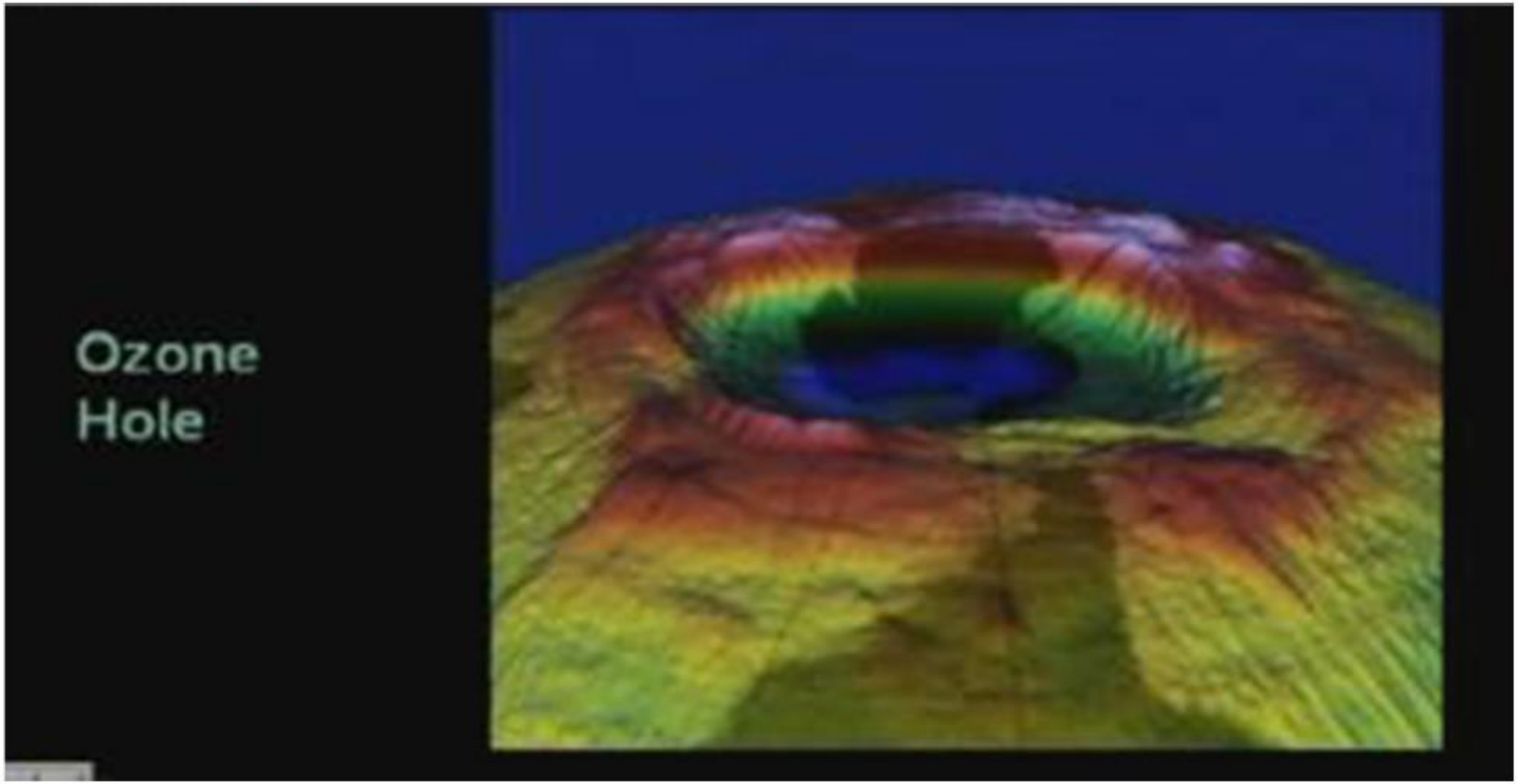
Weather Forecasting

Hurricane over
Dennis 1990

Astronomy



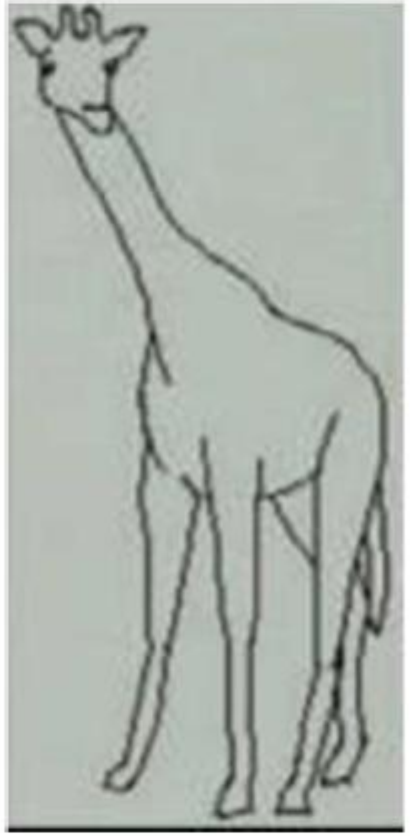
Astronomy



Automated Inspection

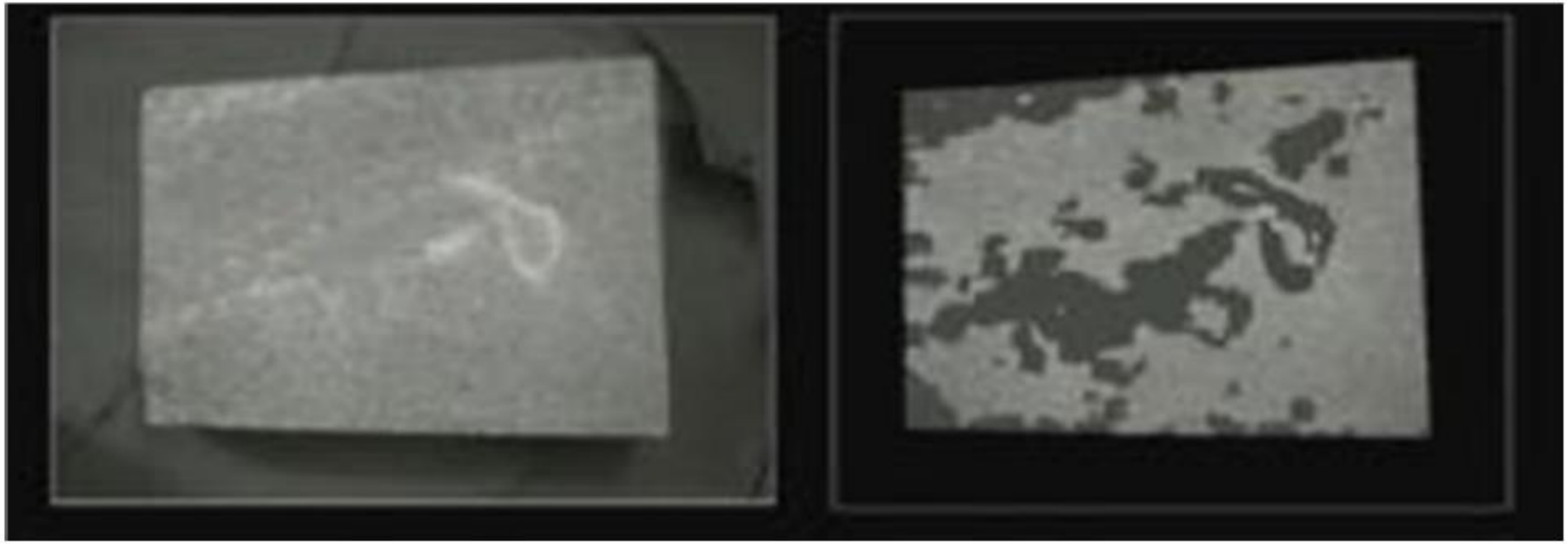
Bottling plant automation

Automated Inspection



Importance of boundary information

Automated Inspection of Surface

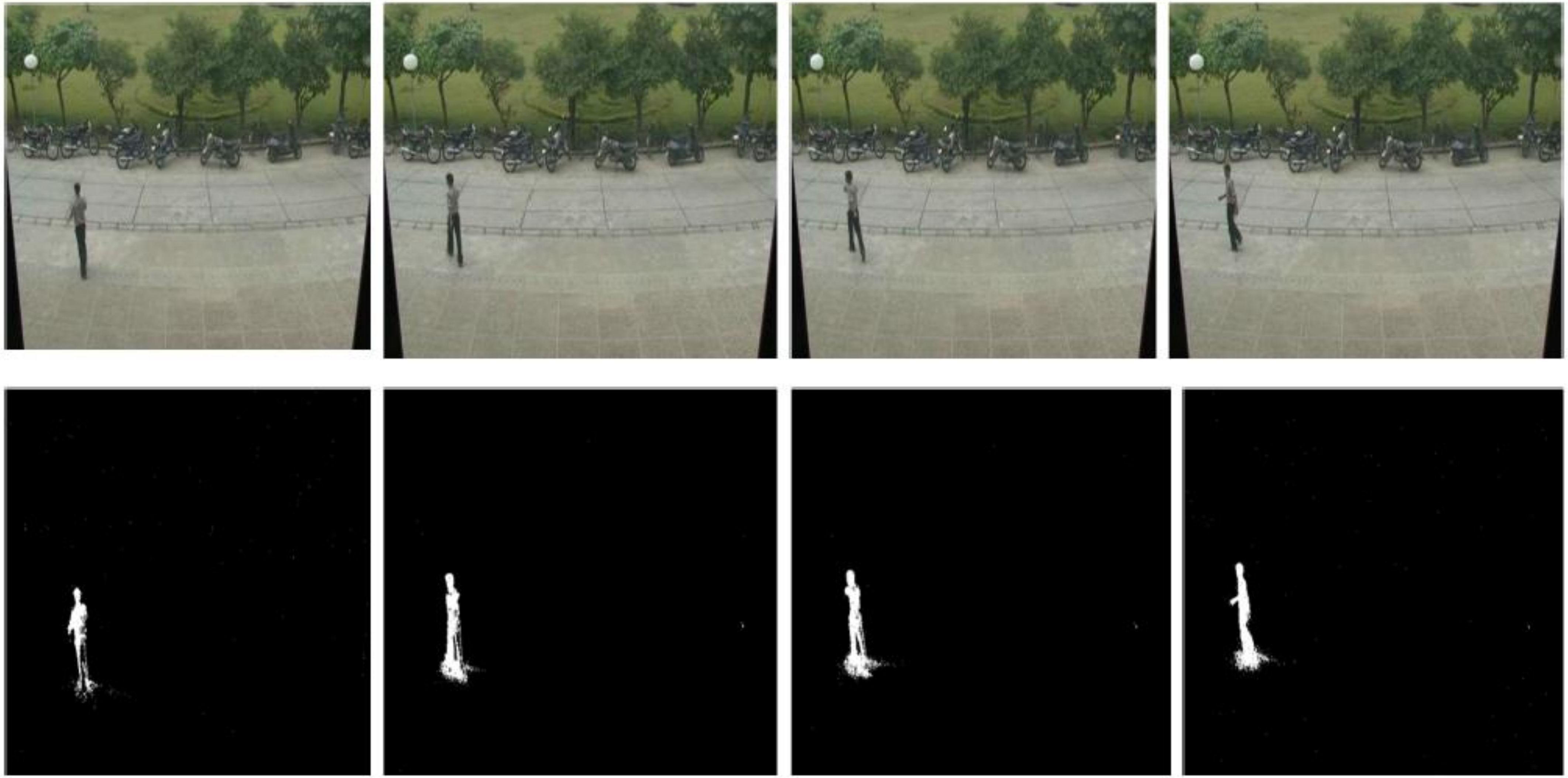


Surface characteristics–texture processing (pattern of surface of object)

Automated Inspection of IC Manufacturing

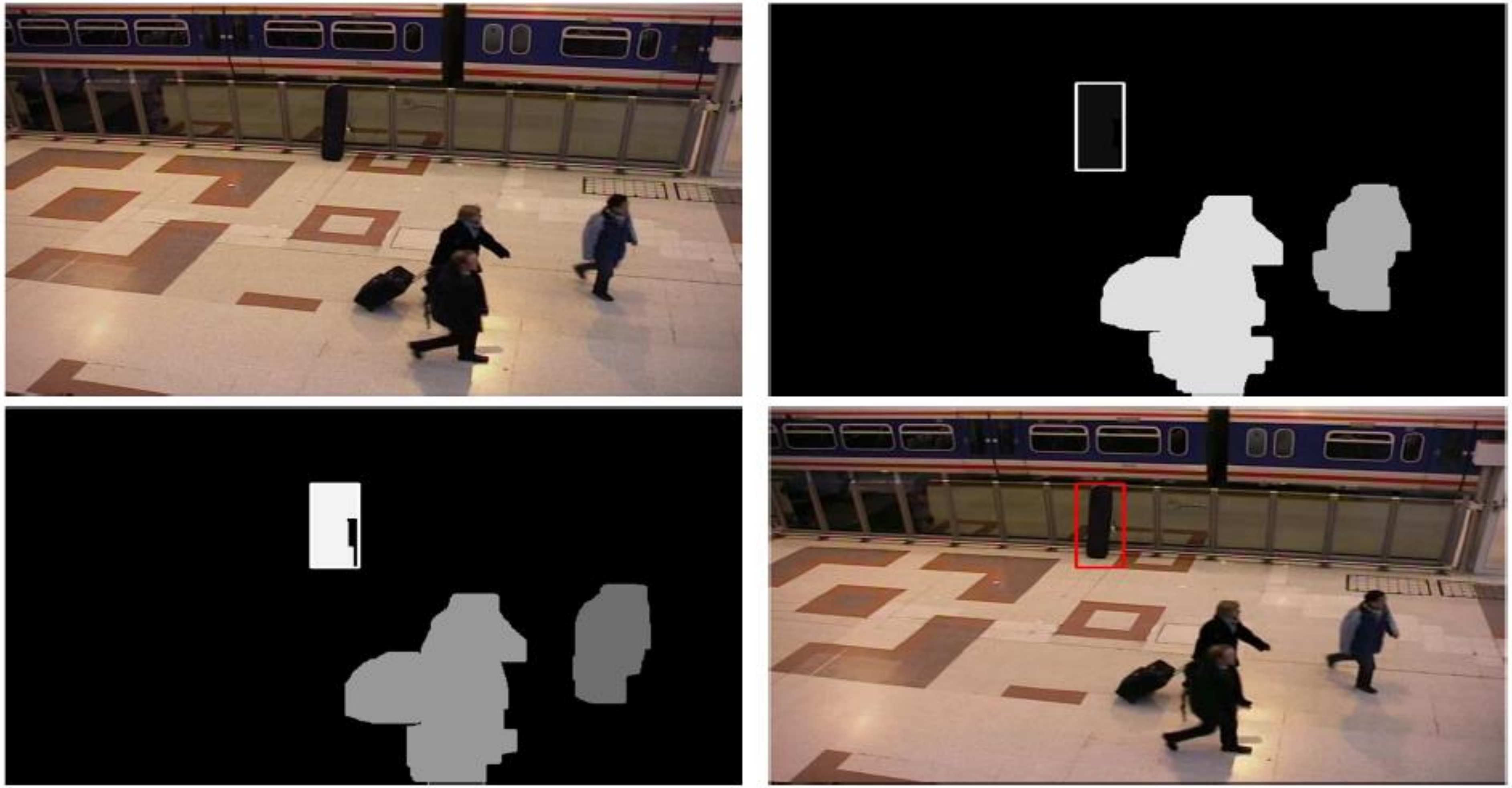
(a) Broken bond (b) Missing bond

Movement Detection



Movement Detection

Static and dynamic object Detection



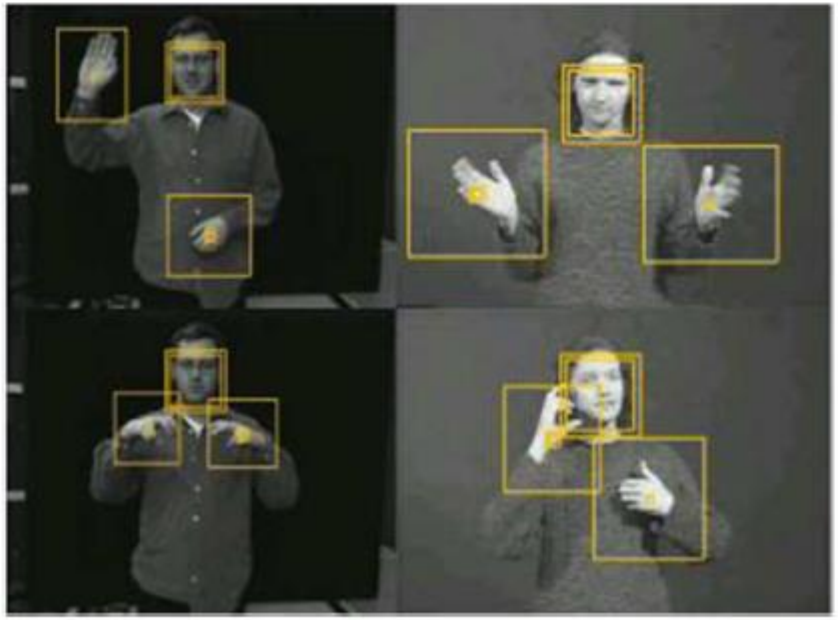
Abandoned Object Detection

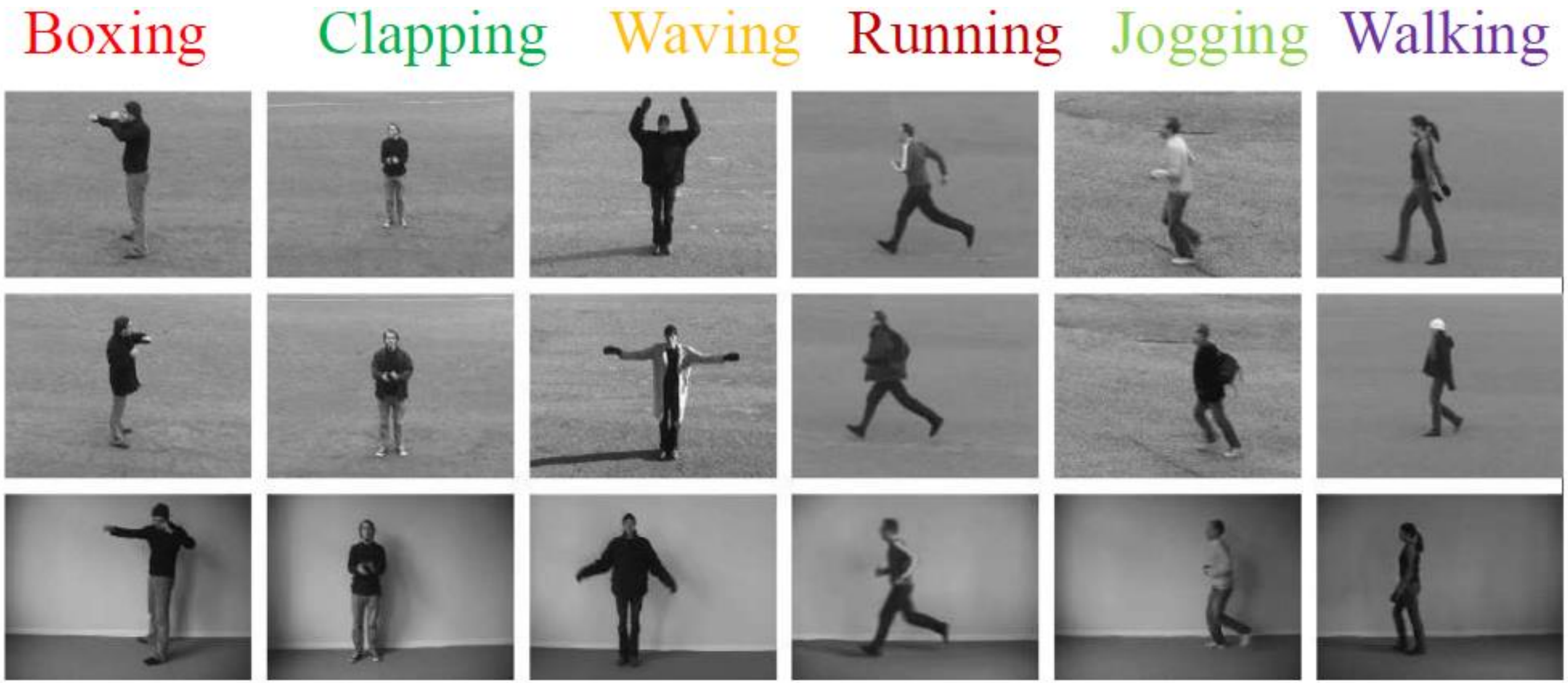
Fingerprint detection and object detection

- Number plate recognition for speed cameras/ automated toll systems
- Fingerprint recognition
- Enhancement of CCTV images

To make human computer interfaces more natural

- Face recognition
- Gesture recognition




Boxing
Clapping
Waving
Running
Jogging
Walking

Human Activity Recognition

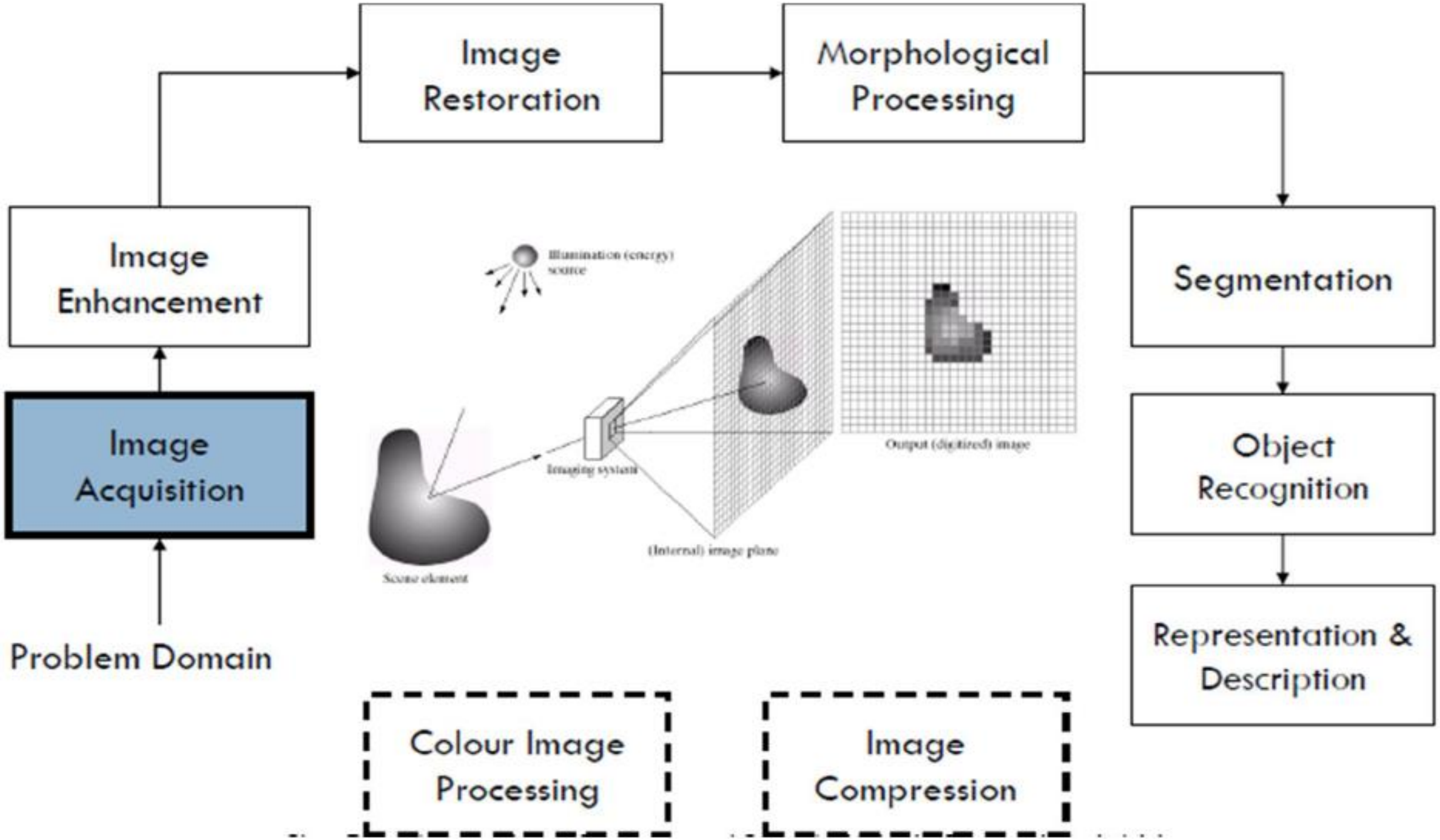


Image Restoration
Morphological Processing
Image Enhancement
Segmentation
Image Acquisition
Object Recognition
Problem Domain
Representation & Description
Colour Image Processing
Image Compression
Illumination (energy) source
Output (digitized) image
Imaging system
(Internal) image plane
Scene element

Image Enhancement

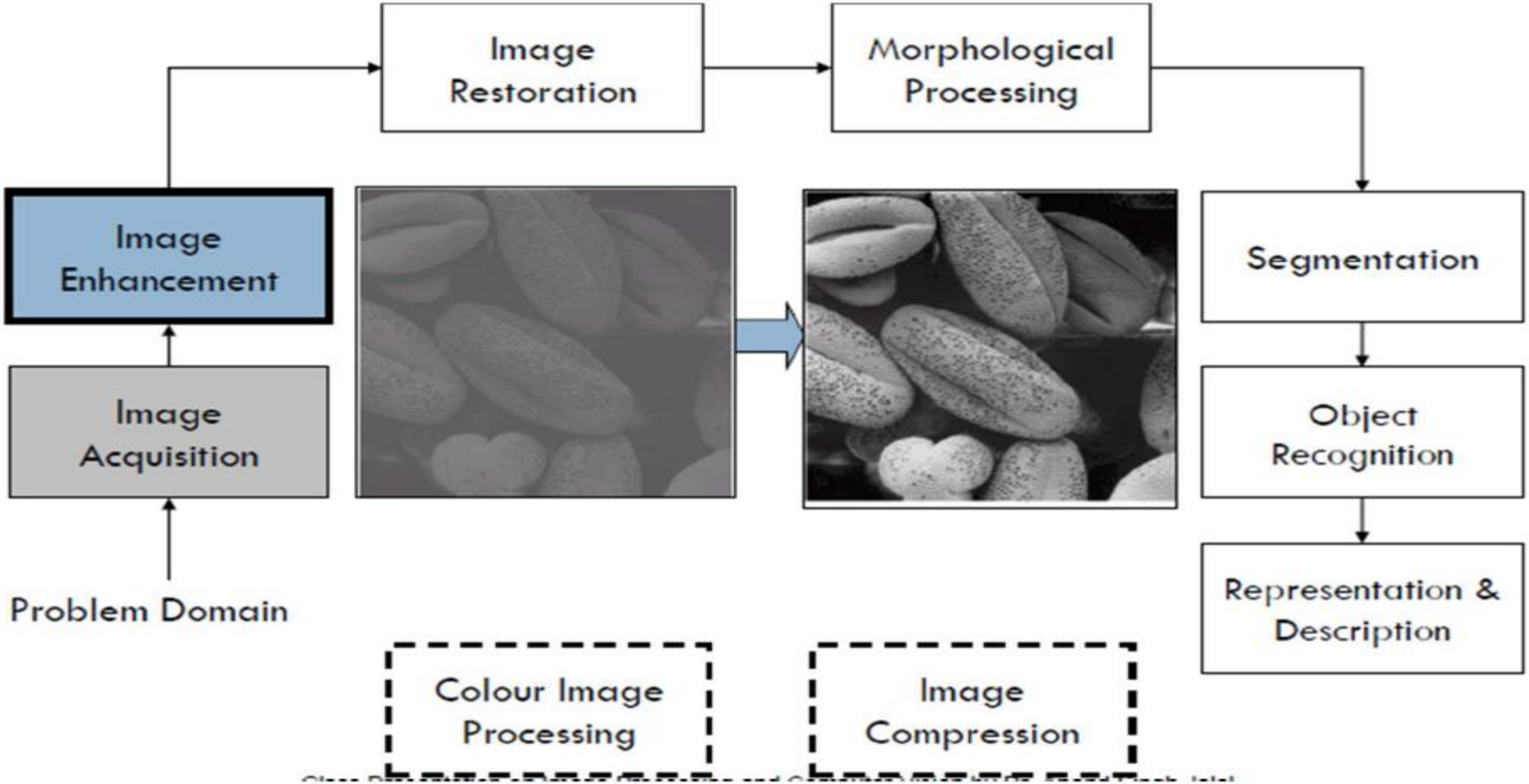
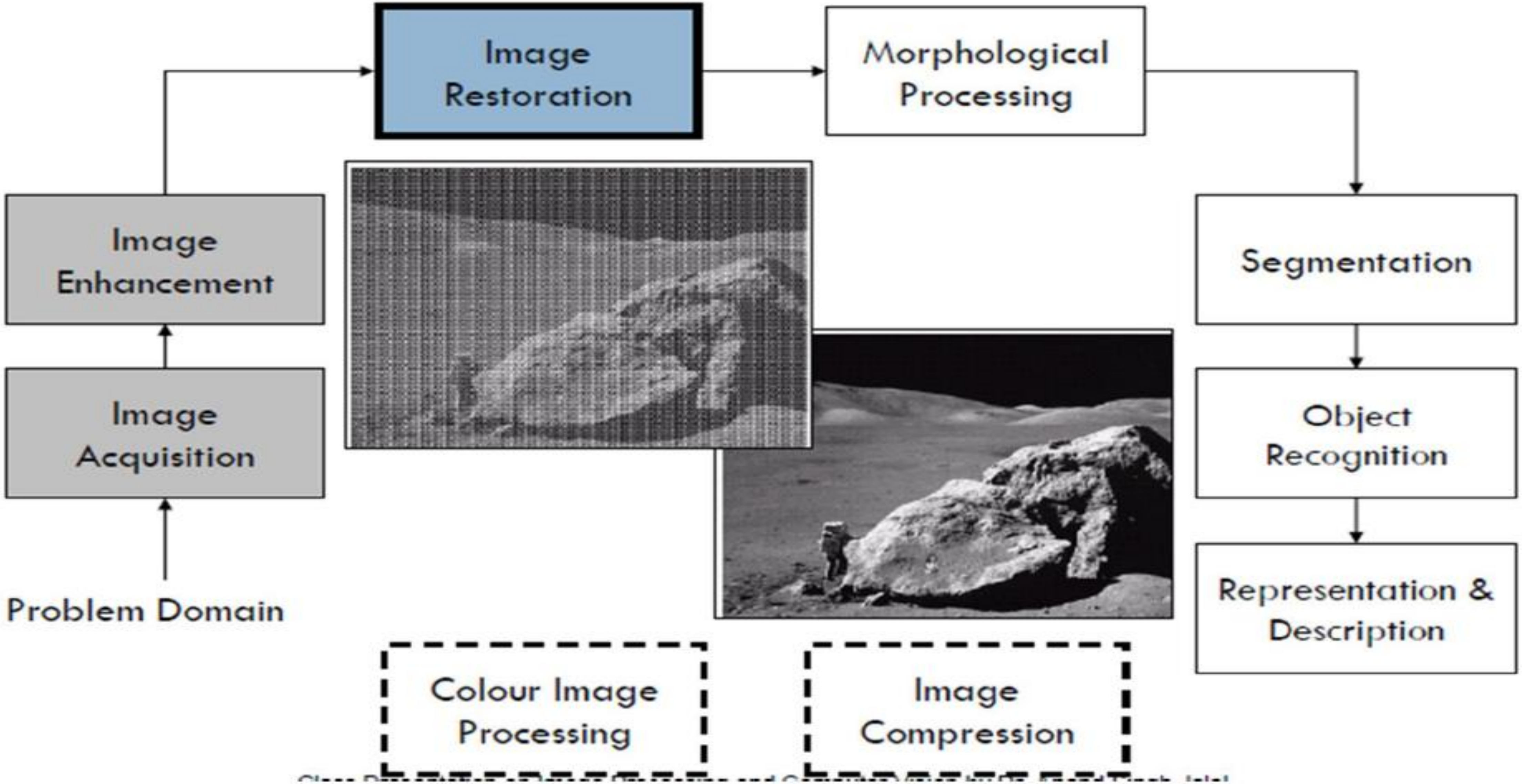
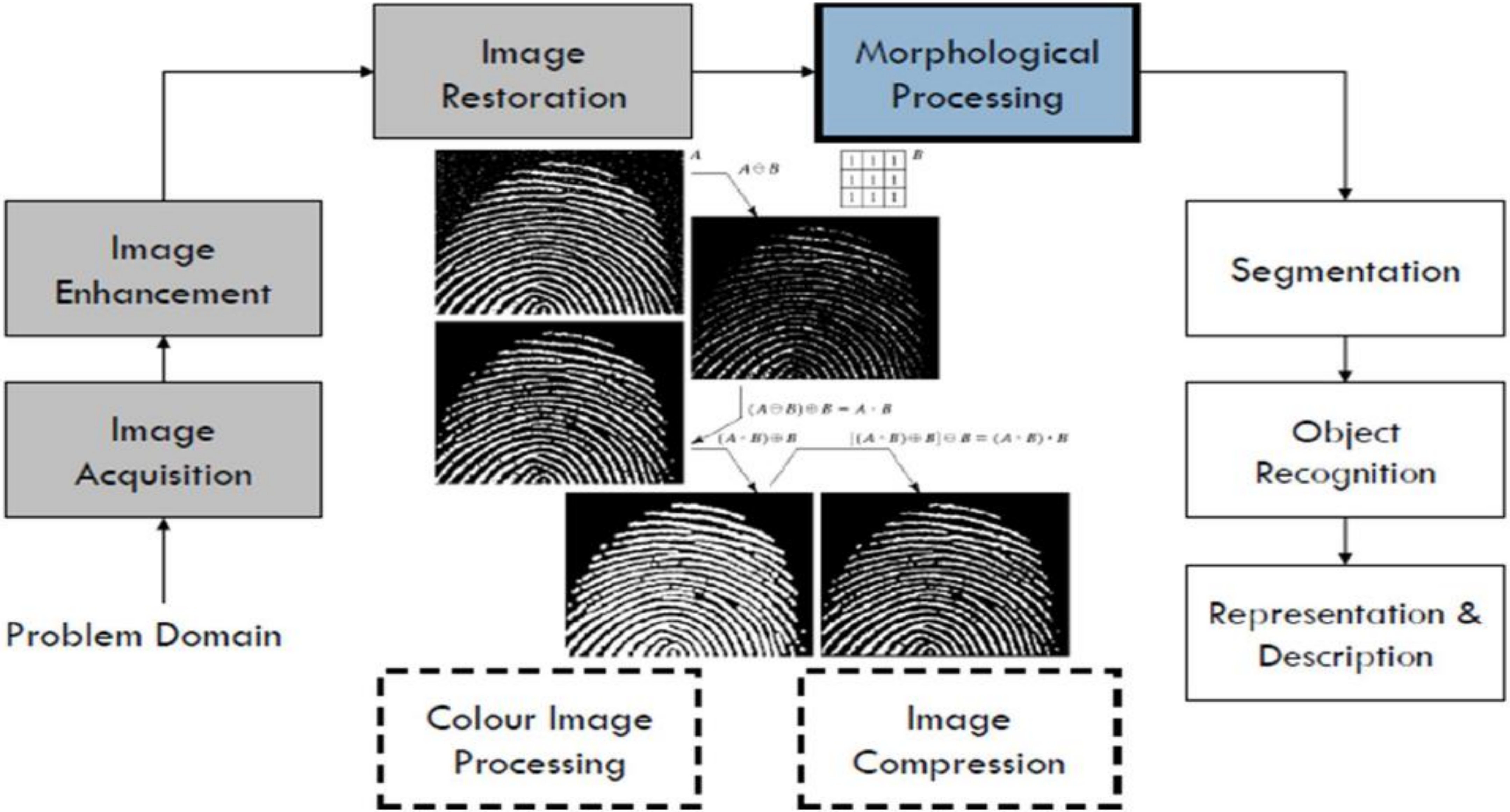


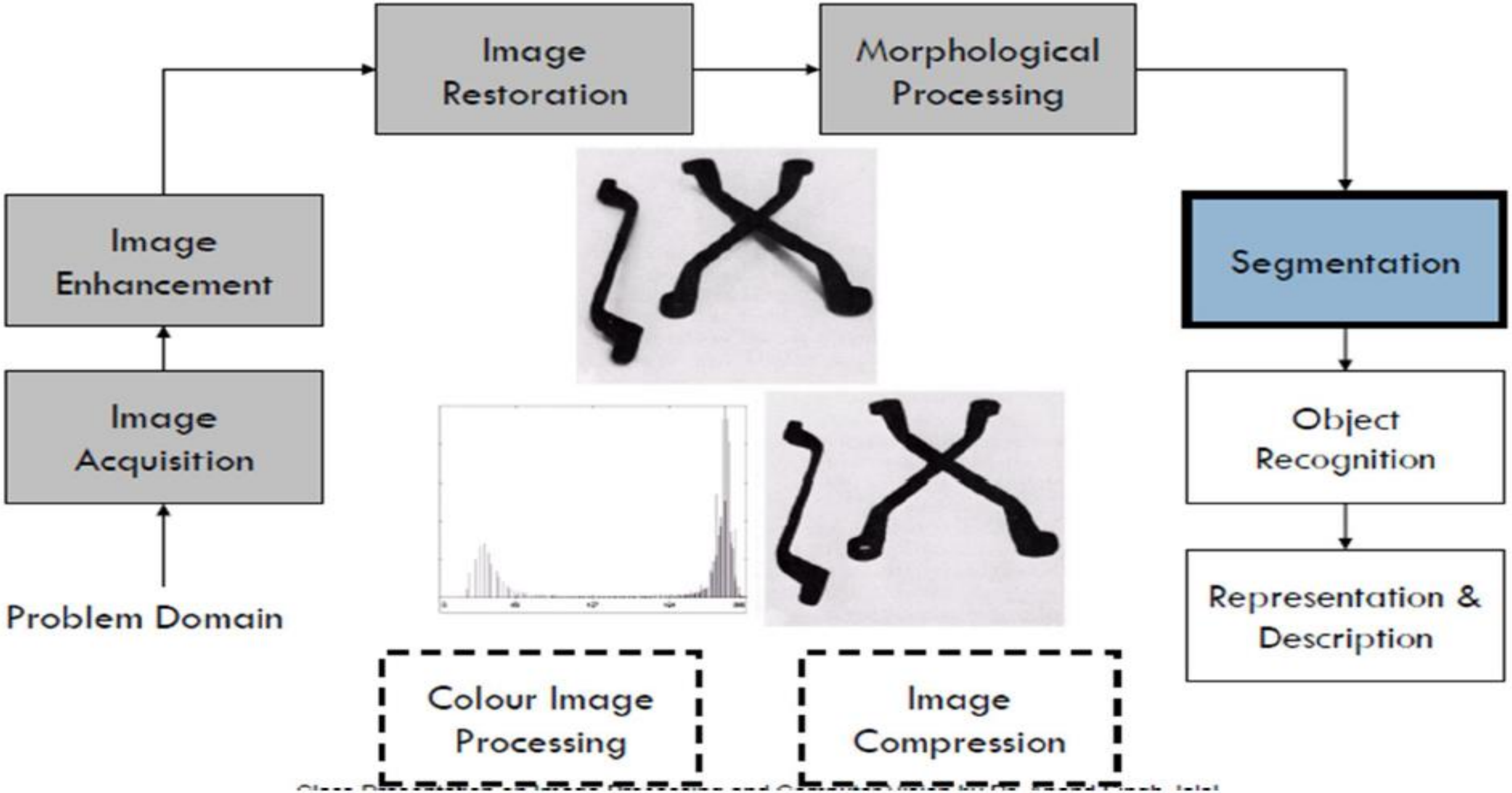
Image Restoration



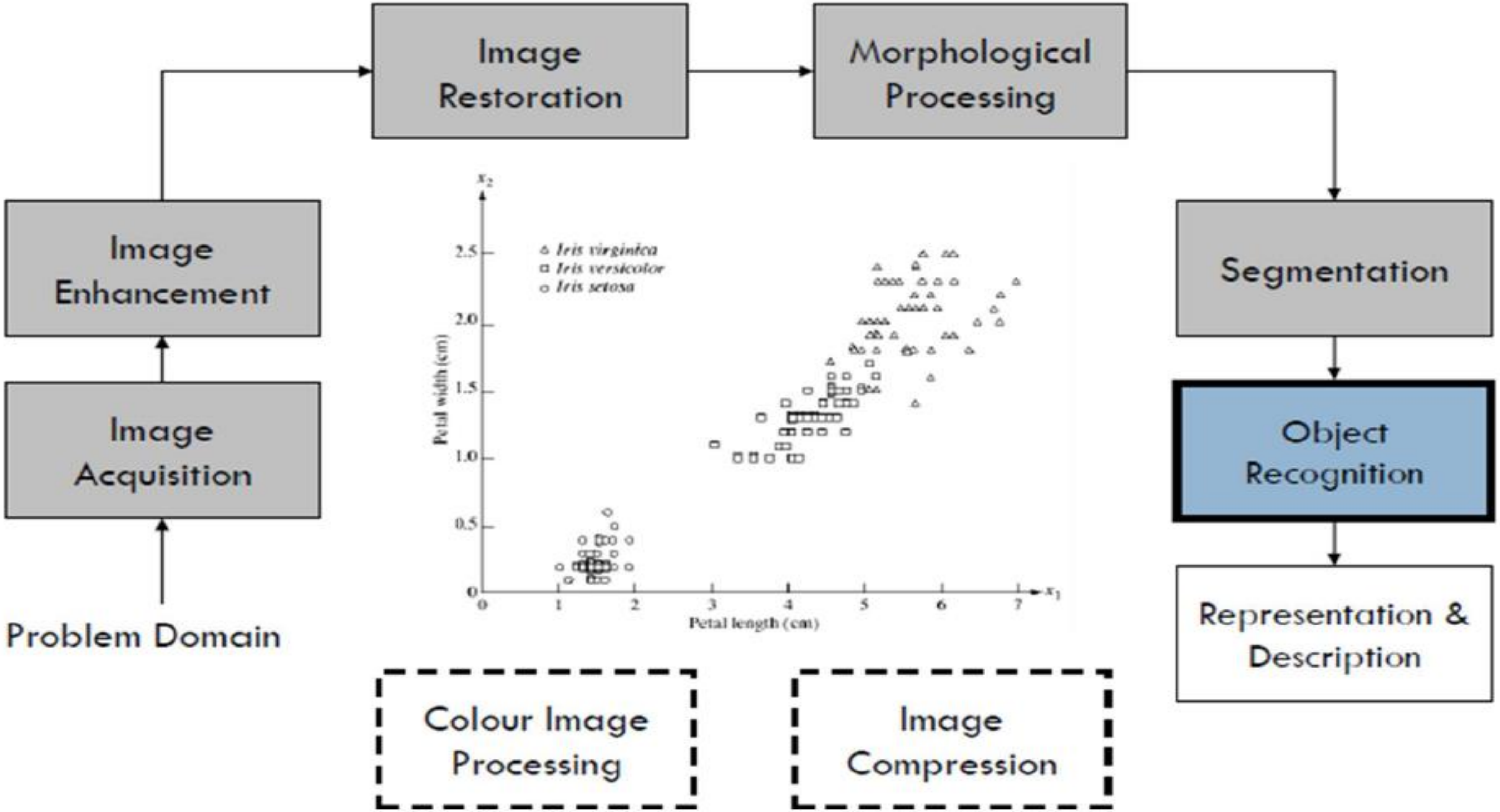
Morphological Processing



Segmentation



Object Recognition



Representation and Description

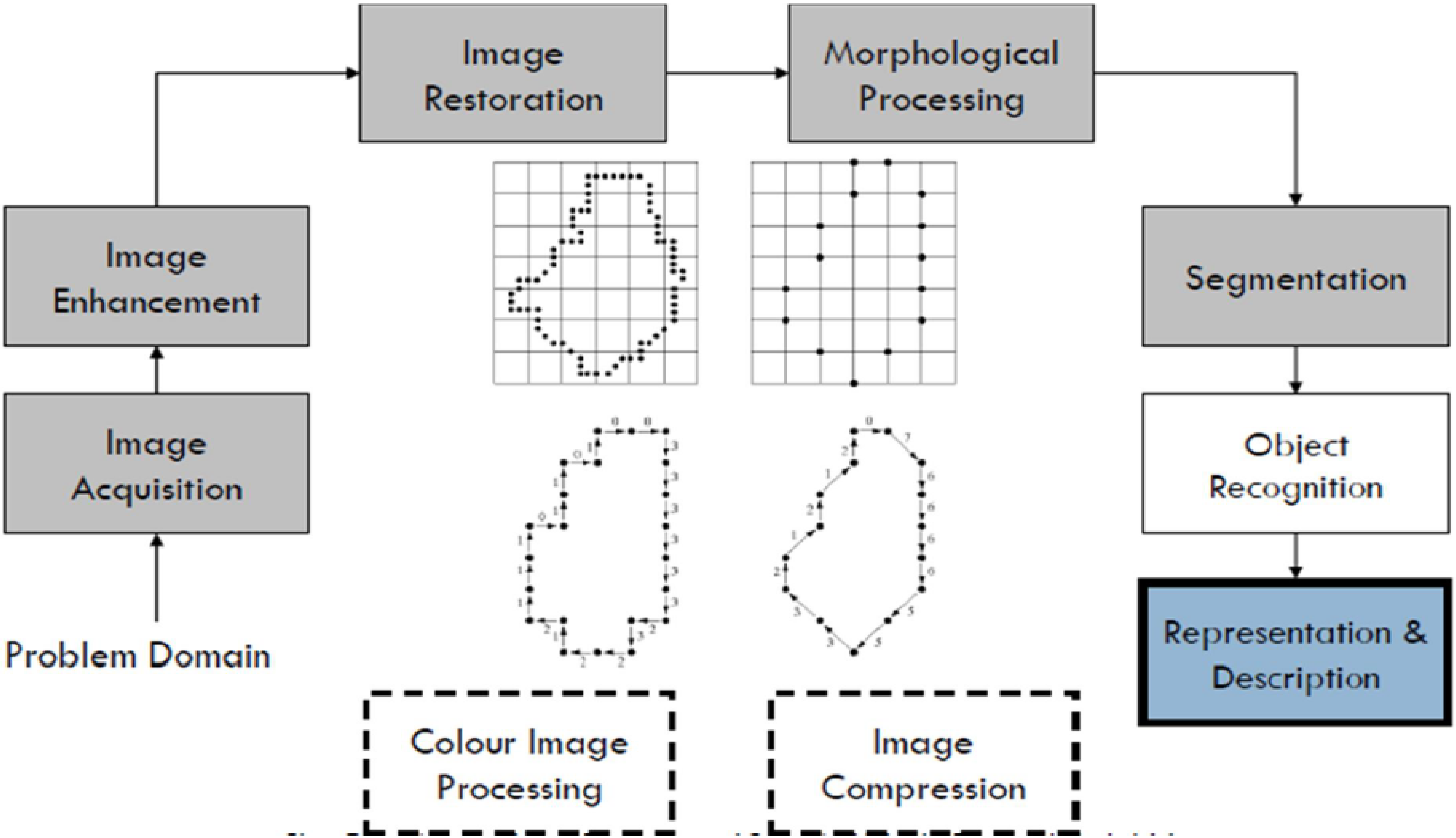
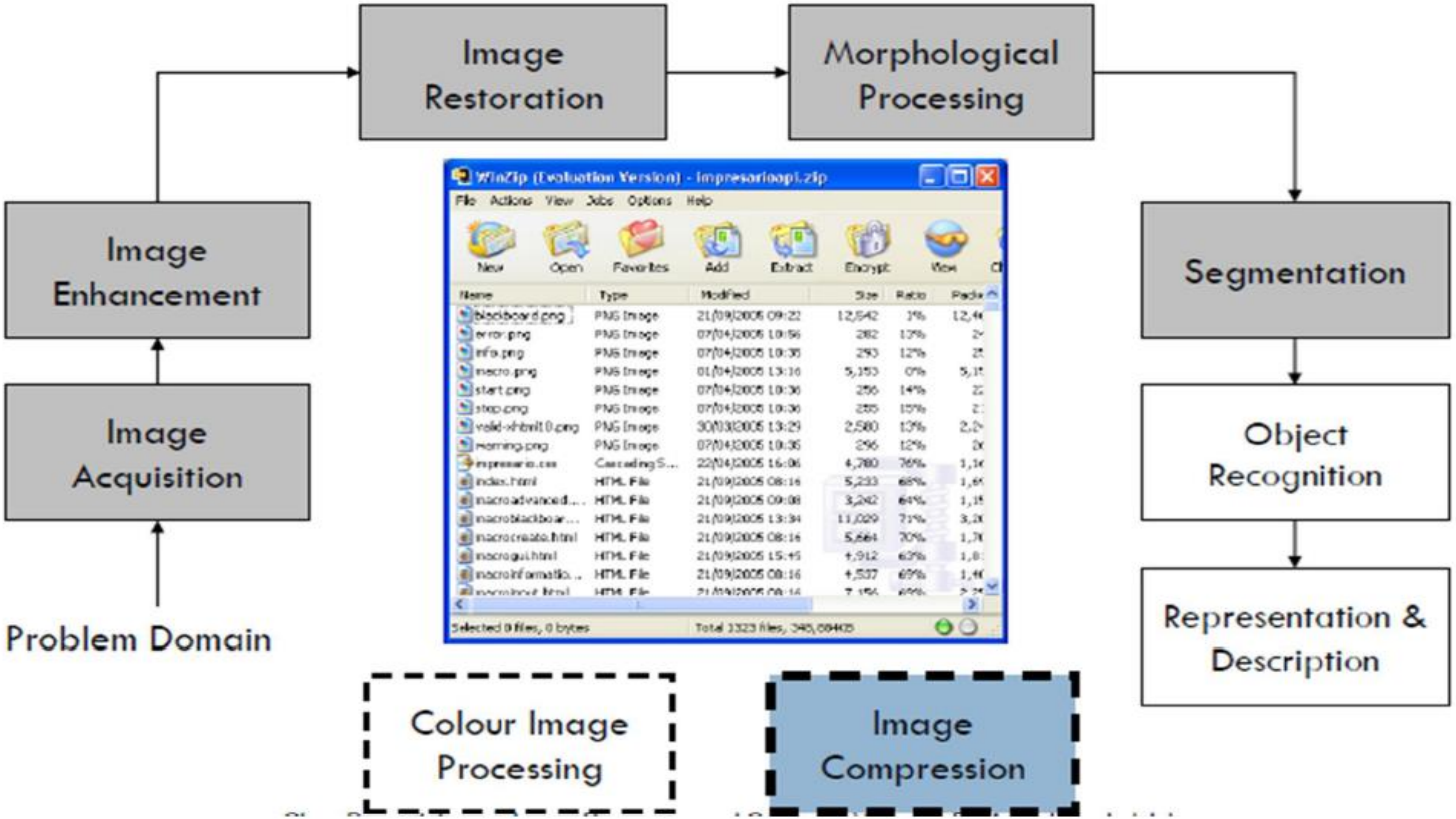
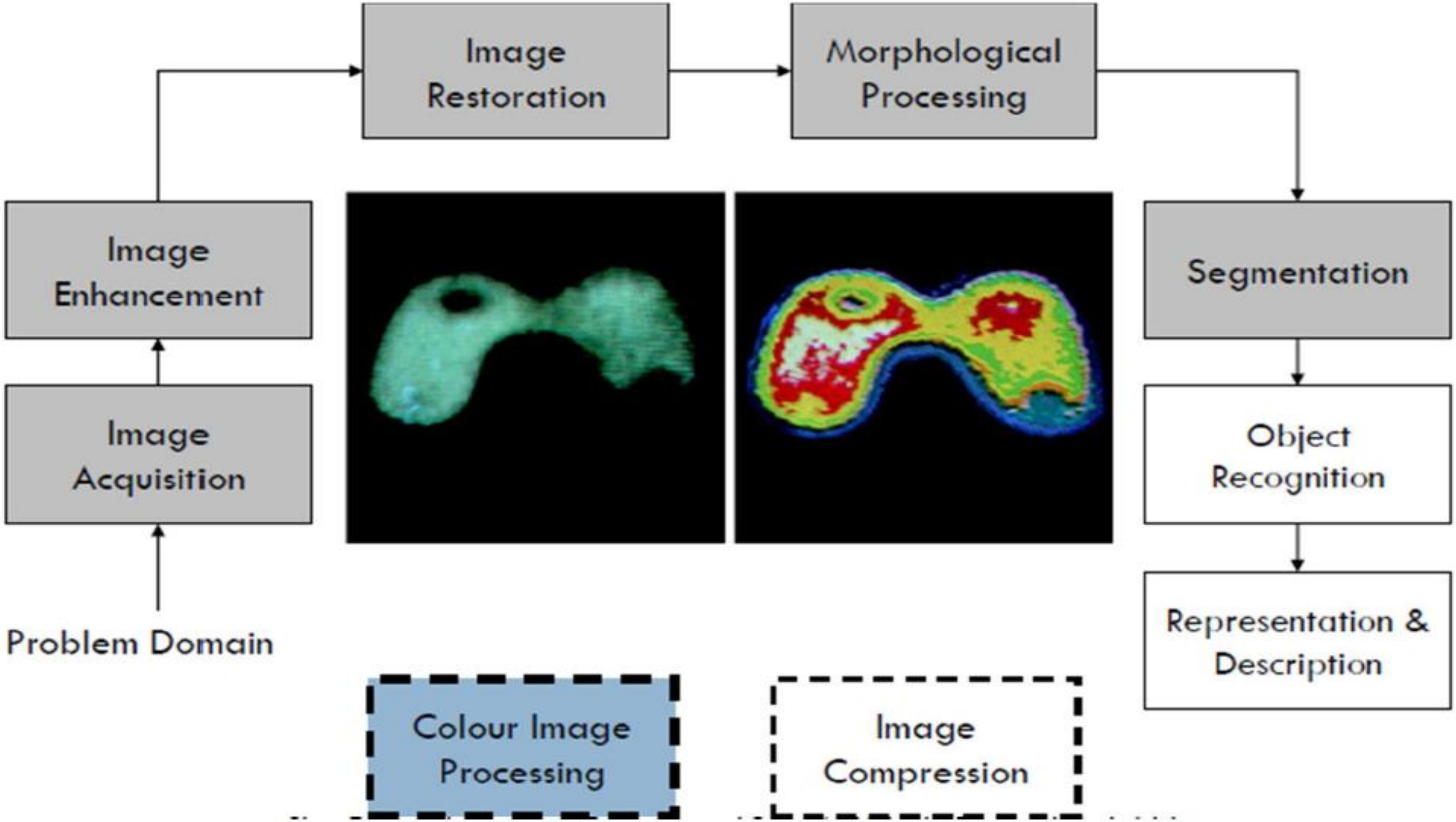


Image Compression



Color Image Processing



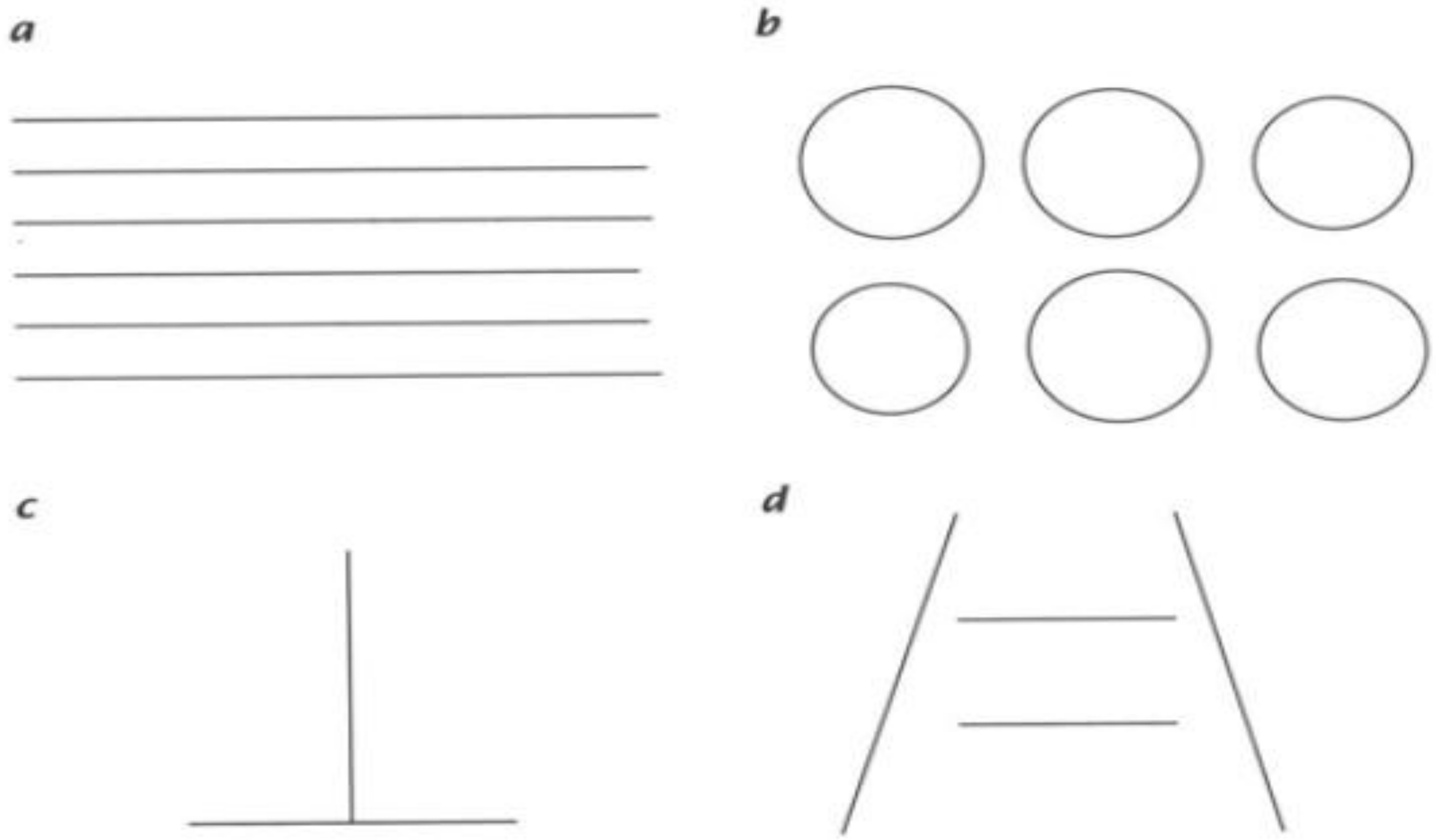
Human and Computer Vision

- We can't think of image processing without considering the human vision system.
- We observe and evaluate the images that we process with our visual system.

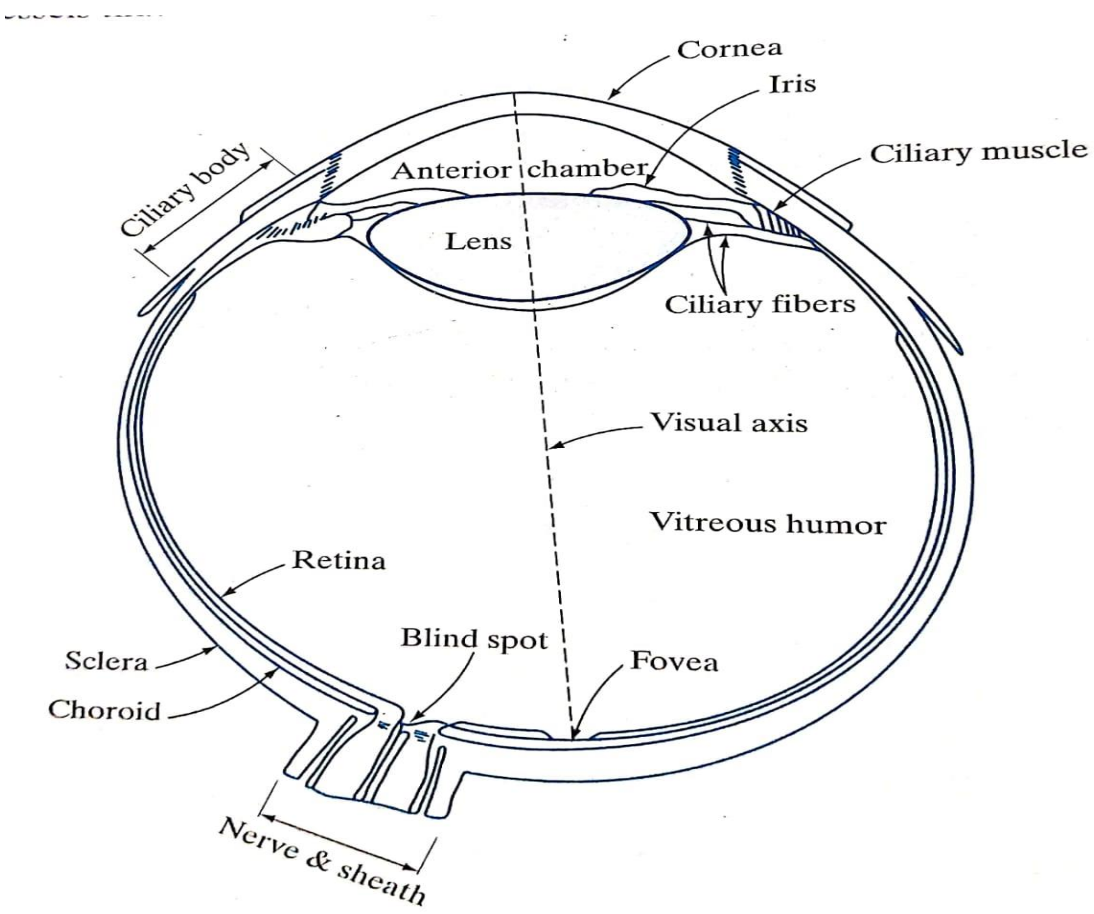
Simple questions

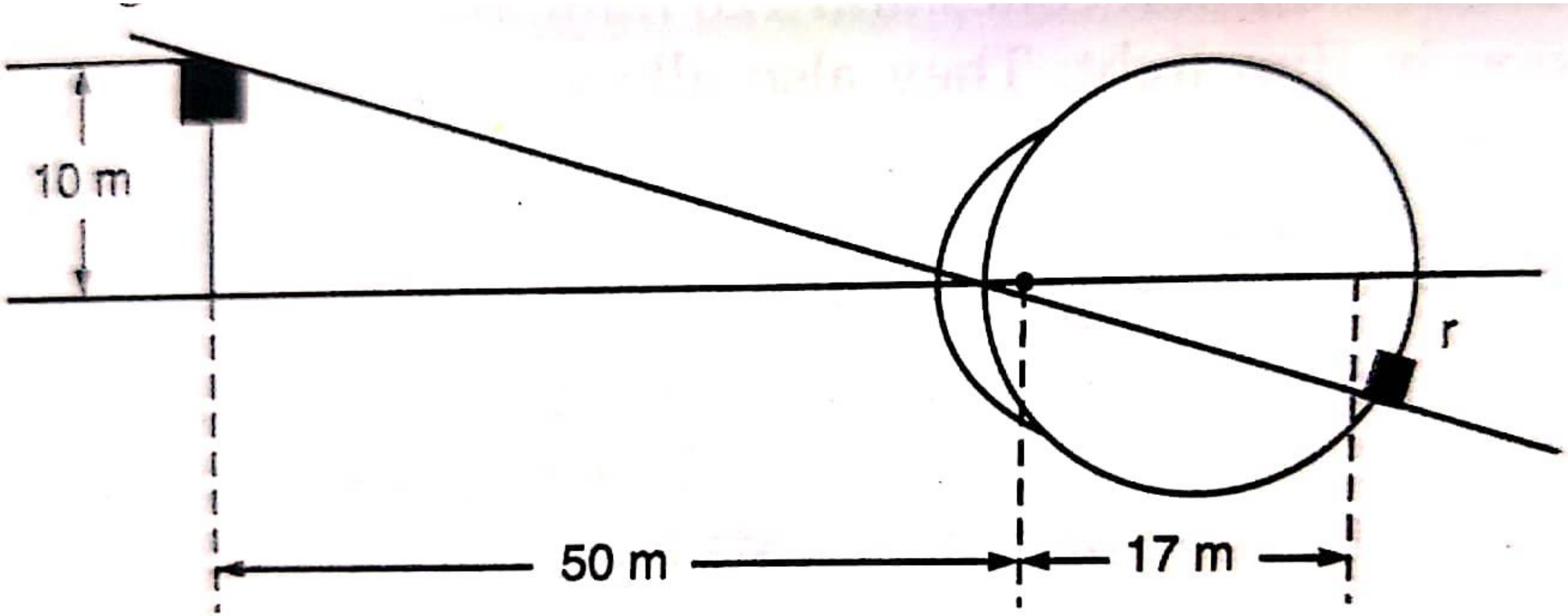
- What intensity differences can we distinguish?
- What is the spatial resolution of our eye?
- How accurately we estimate and compare distances and areas?
- How do we sense colors?
- By which features can we detect and distinguish objects?

Test images



Structure of an Human eye




10 m
r
50 m
17 m

Lens & Retina

- Lens

  both infrared and ultraviolet light are absorbed, in excessive amounts, can cause damage to the eye.

- Retina

  Innermost membrane of the eye. When the eye is properly focused, light from an object outside the eye is imaged on the retina.

Receptors

- Receptors are divided into 2 classes:
  - Cones
  - Rods

Cones

- 6-7 million, located primarily in the central portion of the retina (muscles controlling the eye rotate the eyeball until the image falls on the fovea).
- Highly sensitive to color.
- Each is connected to its own nerve end thus human can resolve fine details.
- Cone vision is called photopic or bright-light vision.

Rods

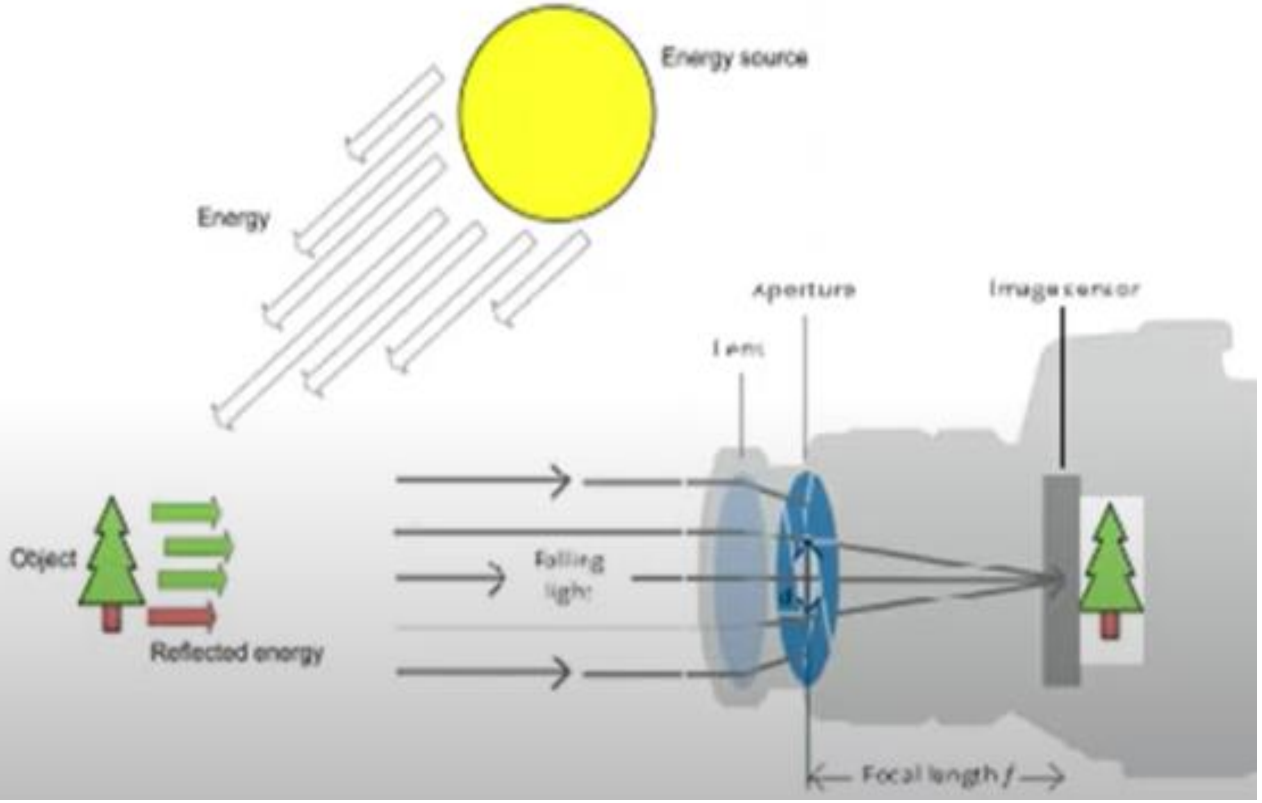
- 75-150 million, distributed over the retina surface.
- Several rods are connected to a single nerve end reduce the amount of detail discernible.
- Serve to give a general, overall picture of the field of view.
- Sensitive to low levels of illumination.
- Rod vision is called scotopic or dim-light vision.

Image Sensing & Acquisition

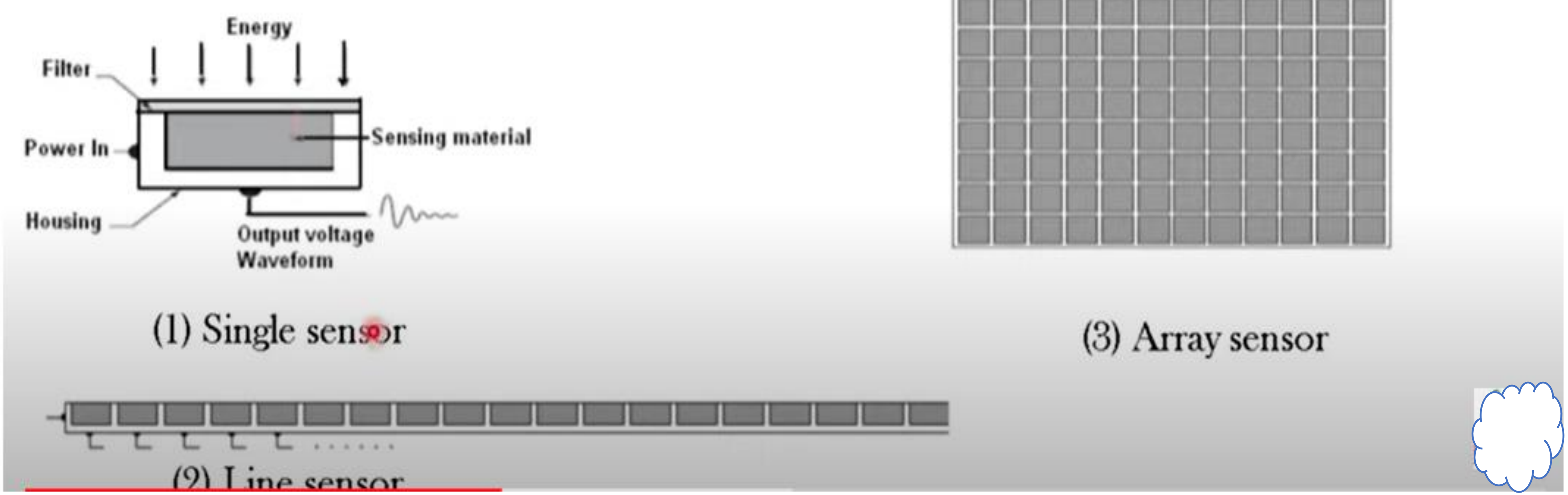
- Image sensing is done by a sensor, which will convert the optical energy into electrical energy or digital image.

In image acquisition there are three components:

(1) Illumination
(2) Optical system (lens system)
(3) Sensor system



- The three principal sensor arrangements used to transform illumination energy into digital images are:
  (1) Single sensor
  (2) Line sensor
  (3) Array sensor



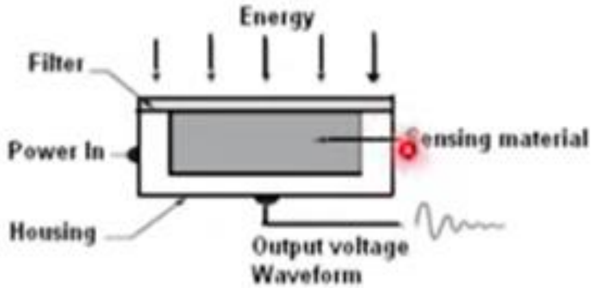
(1) Single sensor

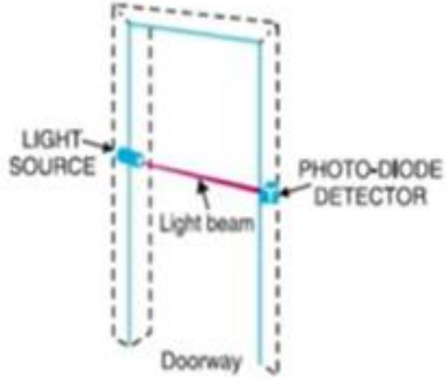
(3) Array sensor

(2) Line sensor

Image Acquisition using a Single Sensor:

- The best example for a single sensor is the photodiode, which is constructed of silicon material whose output voltage waveform is proportional to incident light.
- The use of a filter in front of a sensor improves selectivity. For example, a green (pass) filter in front of a light sensor allows only green light .
- In order to generate a 2-D image using a single sensor, there has to be relative displacements in both the x- and y-directions between the sensor and the area to be imaged.





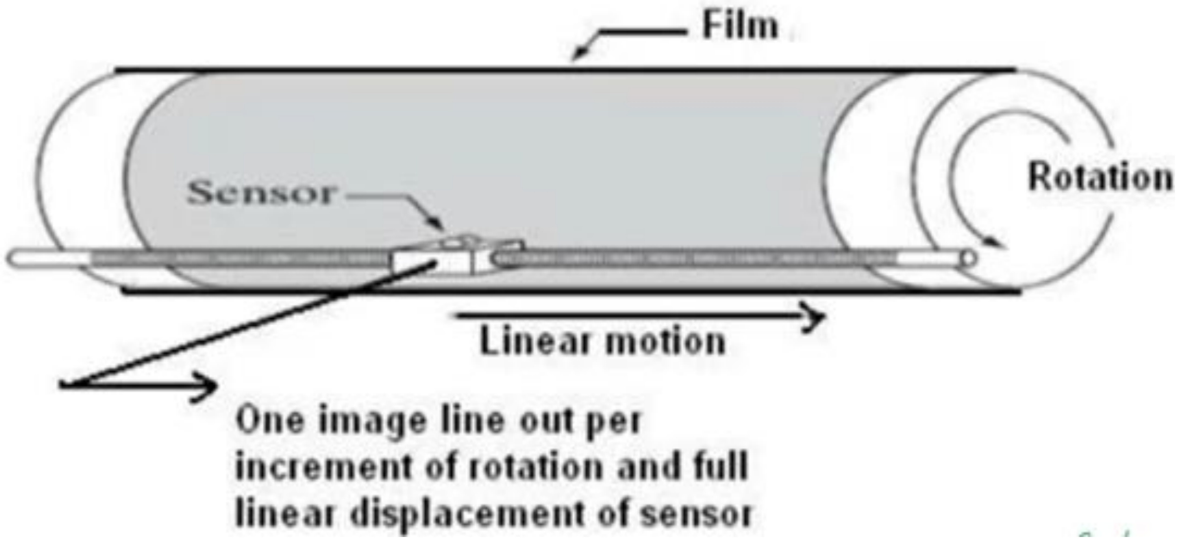
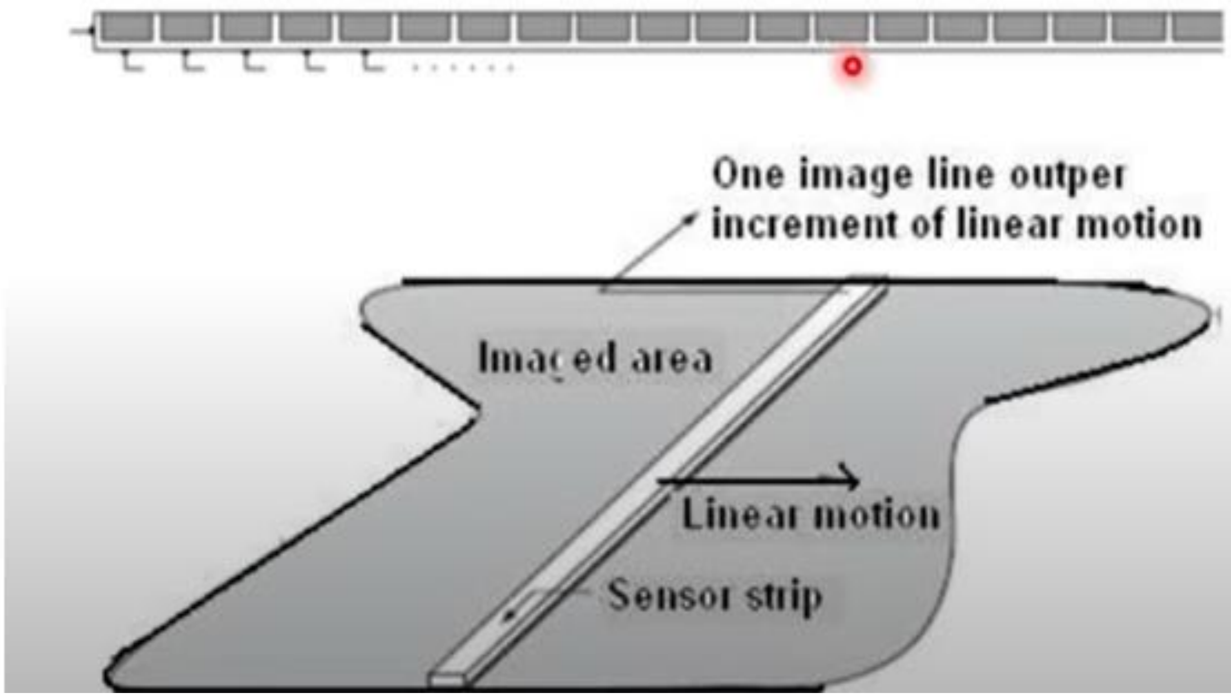


Image Acquisition Using Sensor Strips:

- The sensor strip provides imaging elements in one direction. Motion perpendi to the strips provides imaging in the other direction.
- This arrangement is used in most flat-bed scanners
- Sensing devices with 4000 or more in-line sensors are possible.



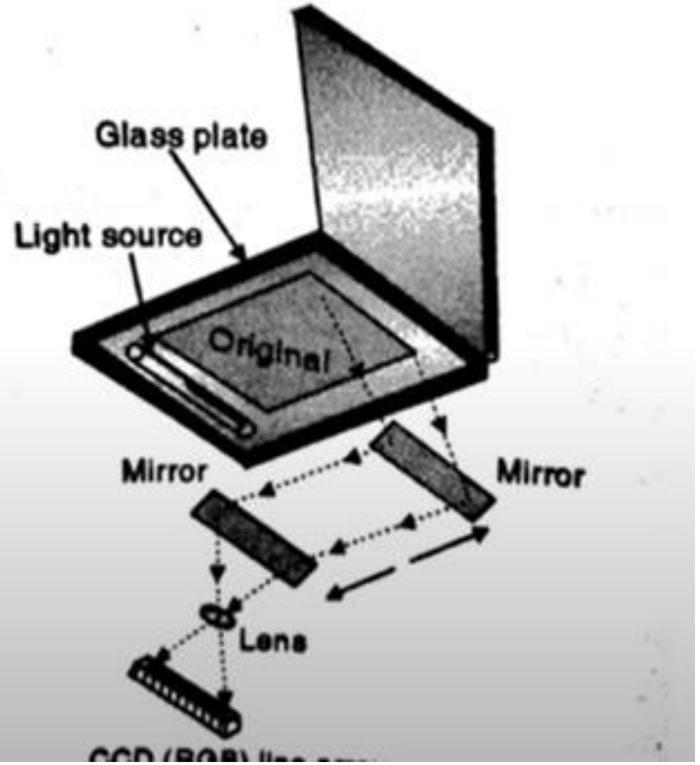
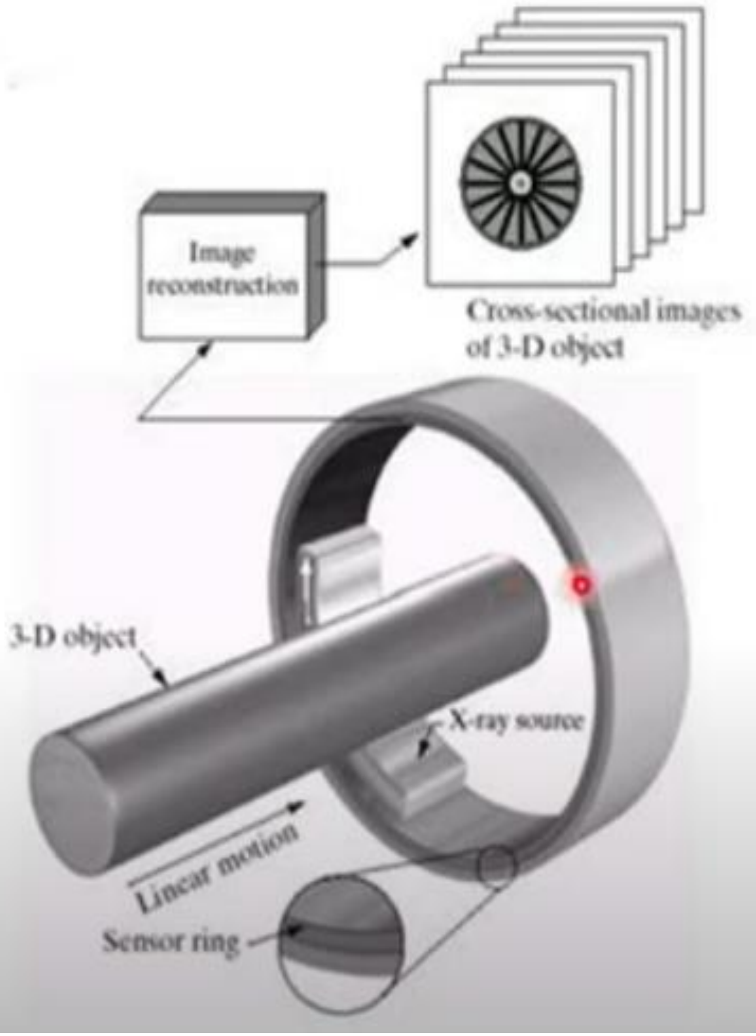


Image Acquisition Using Circular sensor strip

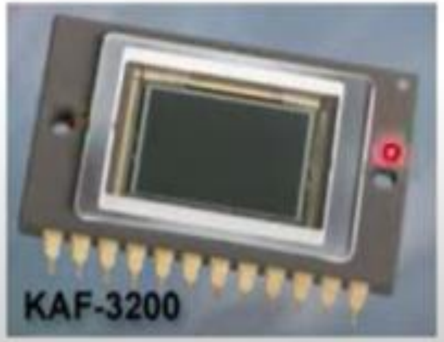


- The sensor strip arrangement mounted in a ring configuration is used in X-ray, medical and industrial imaging to obtain cross-sectional (slice) images of 3-D objects.
- A rotating X-ray source provides illumination and the portion of the sensors opposite the source collect the X-ray energy that pass through the object.

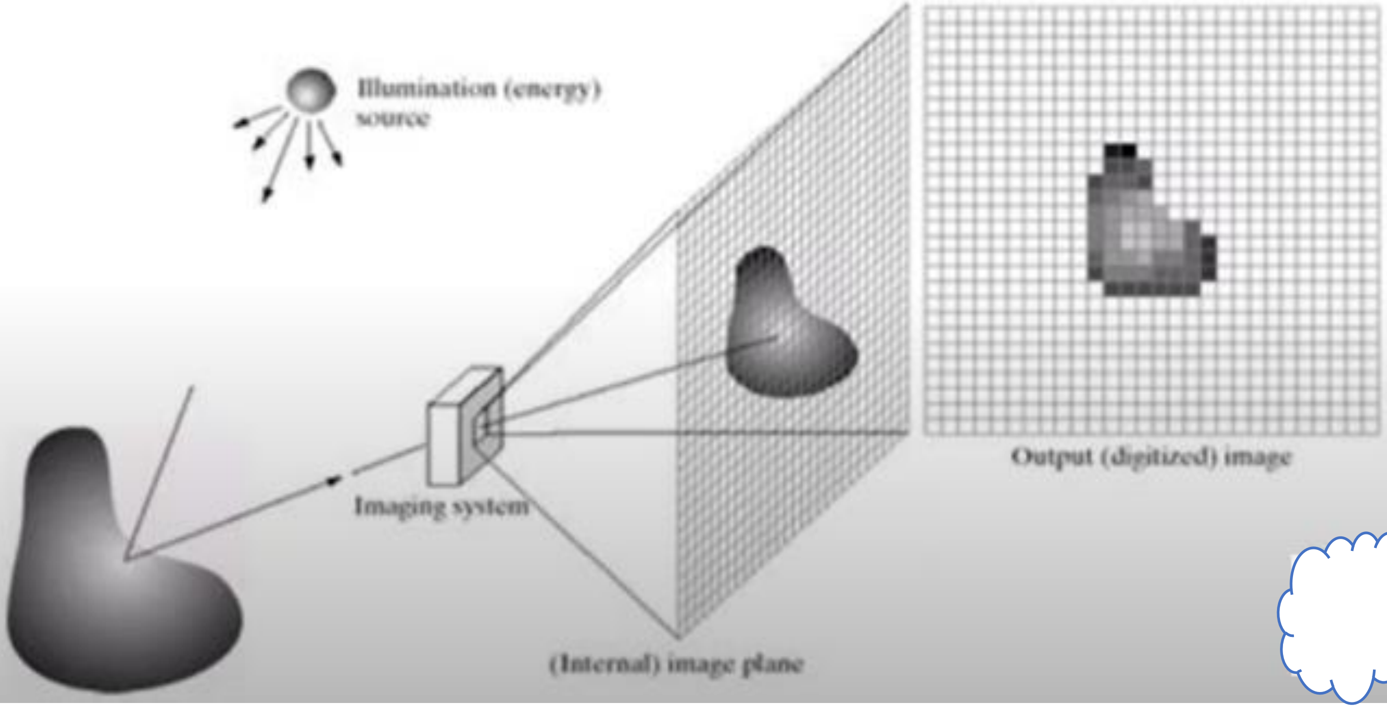
During a computerized tomography (CT) scan, a thin x-ray beam rotates around an area of the body, generating a 3-D image of the internal structures

Image Acquisition using Array sensor:

- The typical array sensor is a CCD array, which can be manufacture with 4000*4000 elements or more.
- The response of each sensor is proportional to the integral of the light energy projected onto the surface of the sensor.



CCD KAF-3200E from Kodak.
(2184 x 1472 pixels,



A Simple Image formation model:

- An image is defined by two dimensional function f(x,y). The value or amplitude of f at spatial coordinates (x,y) is a positive scalar quantity.

  $$0 < f(x,y) < \infty$$

- The function f(x,y) may be characterized by two components: (1) the amount of source illumination incident on the scene being viewed and (2) the amount of illumination reflected by the objects in the scene.

  $$f(x,y) = i(x,y)\, r(x,y), \text{ where } 0 < i(x,y) < \infty \text{ and } 0 < r(x,y) < 1$$

- The interval of l ranges from [0,L-1]. Where $l=0$ indicates black and $l=1$ indicates white. All the intermediate values are shades of gray varying from black to white.

Image Sampling and Quantization

- For numerous ways to acquire images, objective is same:
  To generate digital images from sensed data.
- The output of most sensors is a continuous voltage waveform whose amplitude and spatial behavior are related to the physical phenomenon being sensed.
- To create a digital image, we need to convert the continuous sensed data into digital form.
- This involves two processes: sampling and quantization.

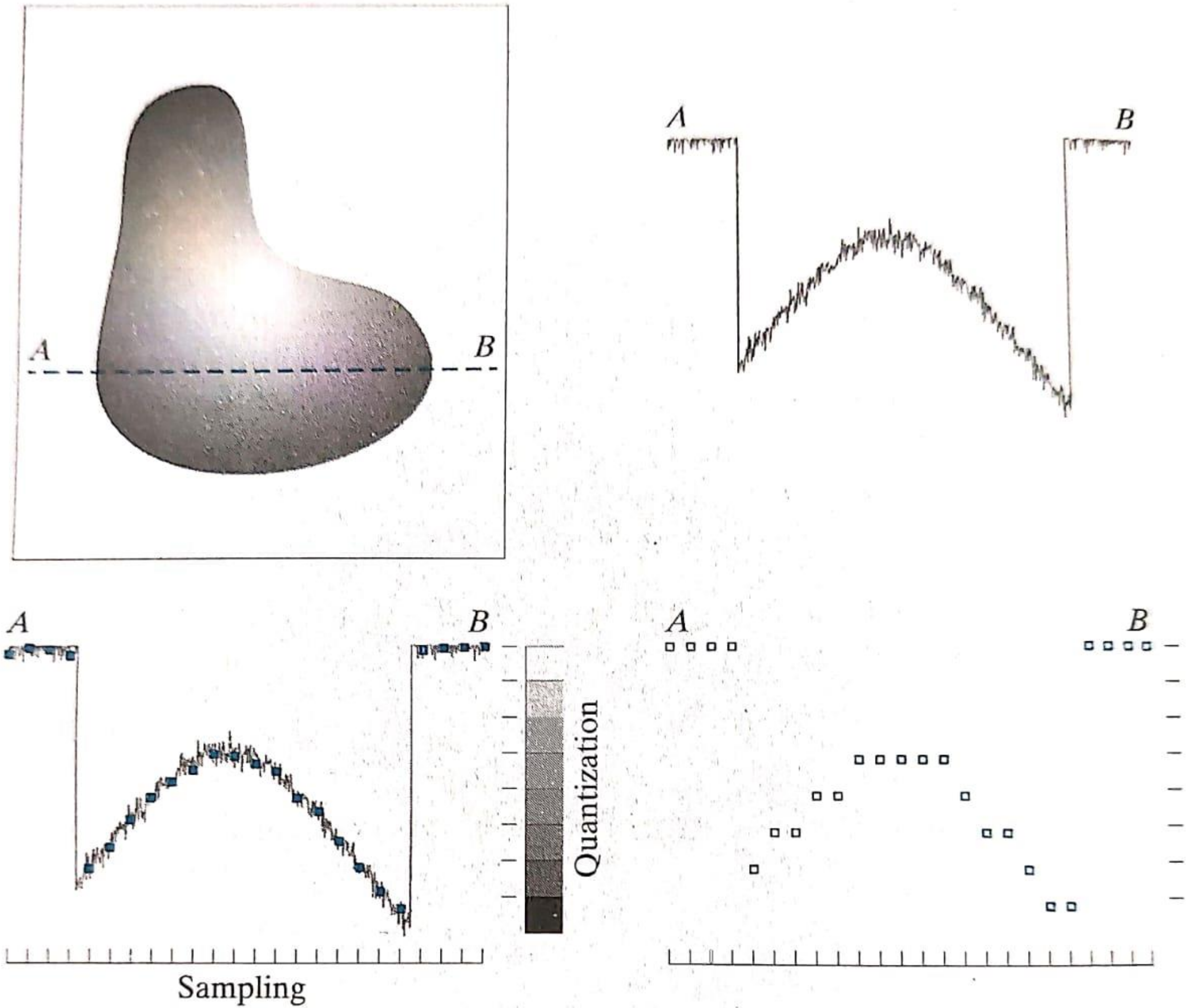
- Sampling the analog signal mean instantaneously measuring the voltage of the signal at fixed interval in time.

- The value of the voltage at each instant is converted into a number and stored.

- The number represents the brightness of the image at that point.

- The "grabbed" image is now a digital image and can be accessed as a two dimensional array of data.
- Each data point is called a pixel (picture element).
- The notation used to express a digital image: I(r,c)
- I(r,c) = The brightness of the image at point (r,c)

- Let a continuous image f is to be converted to digital form.
- An image may be continuous with respect to the x and y coordinates, and also in amplitude.
- To convert it to digital form, we have to sample the function in both coordinates and in amplitude.
- Digitizing the coordinate values is called sampling.
- Digitizing the amplitude values is called quantization.



A
B
A
B
A
B
Quantization
A
B
Sampling

- In addition to the number of discrete levels used, the accuracy achieved in quantization is highly dependent on the noise content of the sampled signal.

- In practice, the method of sampling is determined by the sensor arrangement used to generate the image.

- When an image is generated by a single sensing element combined with mechanical motion, the output of the sensor is quantized in the manner described above.

- When a sensing strip is used for image acquisition, the number of sensors in the strip establishes the sampling limitations in one image direction.

- Mechanical motion in the other direction can be controlled.

- When a sensing array is used for image acquisition, there is no motion and the number of sensors in the array establishes the limits of sampling in both directions.